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Abstract

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DFT - Density functional theory

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2.1.1 Spatial weights

Spatial weights, formalized by the seminal work by Cliff & Ord (1973), connect objects in geographic space using the spatial relationship between them. They serve for defining spatial neighbourhoods and geographically relevant sites. These relationships can be distance-based (k-nearest neigbours, distance bands), based on a membership in a geographic group (block weights) or adjacency-based (contiguity). Each pair of observations is assigned a weight, which can either be boolean (1 for a spatial relationship, 0 otherwise), or specific numerical value (e. g. inverse distance) among others. Afterwards, we will further focus on contiguity-based spatial weights, as those can be directly impacted by violating polygonal coverage.

Spatial weights can be represented as a matrix, denoted as W. W is a $N \times N$ matrix, where N is the number observations. Each element w_{ij} corresponds to the weight between observations i and j, that is, how much observation j influences observation i. Because geographic datasets are rarely fully interconnected, most elements in W are 0. To ensure computational and memory efficiency, tools like libpysal (Rey et al., 2022), geoda (Anselin et al., 2006) or spdep (Bivand, 2022) also implement sparse matrix, which stores only non-zero elements.

Spatial weight matrix can be understood as a graph adjacency matrix, therefore spatial graph is another representation of spatial weights. A weighted graph is defined as an ordered triple $G = (V, E, \omega)$, where V is a set of nodes, E is a set of edges representing connectivity between nodes and $\omega : E \to \mathbb{R}$ is a function that assigns real numbers (weights) to edges. Two nodes $i, j \in V$ are adjacent if there exists an edge $\{i, j\} \in E$ connecting them. Each node represents an observation and a weighted edge denotes the relationship between observations. Fundamentally, spatial weights matrices and graphs store the exact same topological information; the matrix is simply the algebraic representation of the spatial graph.

-fig nejaky data, jejich spatial weight matrix, jejich graph

-neco o symetricnosti?

2.1.1.1 Contiguity

Spatial objects that share a common border are considered to be contiguous. However, "sharing a common border" isn't a rigorous definition, leading to several formal interpretations of contiguity. The primary types of contiguity are named analogically to the movements of chess pieces: *Rook*, *Queen*, and *Bishop*.

For a pair of polygons to be considered *Rook* neighbours, the polygons must share an edge (a boundary of non-zero length). *Queen* contiguity is less strict - the polygons are considered neighbours if they share at least one point (vertex or a shared edge). Bishop contiguity, rarely used on its own is strictly defined as polygons sharing only a point, but no edges. Therefore, it represents the logical difference between *Queen* and *Rook* contiguity (*Queen* - *Rook*). Furthermore, these base definitions can be expanded into higher-order contiguity weights (neighbours of neigbours) to model relationships beyond immediate neighbours.

-nejaky deeper popis contigutiy matrix? symetričnost, row standartization?

-In geographical analysis, Rook or Queen contiguity is usually the standard. There is no concensus, which matrices use when, it is mostly experimental as it depends on the character of the data, but it impacts the outcome of the statistic.. nejaka hlubsi resers na specfication of weight matrix?

2.1.2 Spatial autocorrelation

According to Tobler's Firts law of geography "everything is related to everything else, but near things are more related than distant things". This law implies that spatial data show so called spatial autocorrelation. Spatial autocorrelation is a measure, how similar or dissimilar are nearby observations. We have either positive spatial autocorrelation, meaning that observations geographically close have similar values, or negative spatial autocorrelation, where similar values tend to be dispersed and the data resemble a checkerboard pattern.

-fig negative/random/positive spatial autocorrelation

-spatial autocrrelation pozivame tehdy a tehdy

2.1.2.1 Moran's I

To better understand SAC, we should look at morans cactterpolt (see fig), which visulizes the spatial lag of observations. Spatial lag of an observation i is essentially the weighted average of its neigbours defined by W, therefore it can be denoted as

$$y_{sl-i} = \sum_j w_{ij} y_j \tag{1}$$

where w_{ij} is an element of W capturing the spatial relationship between observations i and j and y_j is the centered value of observation j. Common pratice is to row standartize W, so that the sum of every row is exactly 1. In most cases, neigbours are defined by either QUeen or Rokk contiguity. On Moran scatter plot, spatial lag is represented on y-axis and te centred value of an observation on x-axis.

-fig moran scatter plot

Moran sactterplot alredy provides some some information about both global and local spatial aoutocorrelation. First, global SAC indicators describe the overall degree of clustering or dispersion of the whole dataset. The line on moran scatterplot represetns best linear regression fit for the data values and their spatial lag and its slope corresponds to the value of global Morans I (Moran, 1950), most commonly used metric in SAC. It can be written as:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij} z_i z_j}{\sum_i z_i^2} \tag{2}$$

where n is the number of observations, w_{ij} is an element of W and z_i and z_j is the satnatrtized values of observation i and j, respectively. The value of I is from the interval -1 to 1, where values close 1 indicate positive autocorrelation, vlaues closee to -1 inticate negative spatial autocorrelation and values close to 0 indicate spatial randomness.

Looking back at the morans sactterplot, its four quadrants corrspond to different sutiations:

- High-High
- Low-Low
- Low-High
- High-Low

The category where the observation belongs to can be derived just from the signs of centred observation value and its spatial lag, but to check wheter the observation is in the category rightfully or on accident, we need to use LISA and more importantly, calculate its statistical significance. The most commonly used indicator is local Morans I, defined as:

$$I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j \; ; \; m_2 = \frac{\sum_i z_i^2}{n} \tag{3}$$

where z_i and z_j are centered values on observation i and j, respectively, w_{ij} is the spatial weight for a pair of bservation i and j and m_2 is the second moment (variance). If the reulsting value is positive, the obseravtion is either HH or LL cluster. f its negative, it belongs to Lh or HL.

To identify homogenous or heterogenous clusters that didnt happen by spatially random porcess, the values of local morans I arent as important, but their statistical asignificance is. It is usually measured by p-value, with the null hypothesis that the data values are distributed randomlly and dont give any information about its surroundigs. To obatin the p-value, the obseravtion values are permuted in space (usually 999 times, esda default), and LISA is calculated for those permutations. P-value than corresponds to the proportion of permutations that accept the null hypothesis. For the statistic to be cosidred significan, usually $p < 0.05$. In pratice, we ofthen see maps colouring only those significant clusters, to which we reffer as lISA to cluster maps.

-fig lisa cluster map

We see a lot of grey areas on the map as those arent statistically significant as clusters. We see very little HL and LH clusters commpared to HH and LL which is agrees with toblers first law of geogrpahy - on real data, similar values tend to be near each other.

2.1.2.2 Geary C

Geary C is another SAC indicator. Its global version can be written as

$$C = \frac{(n-1)}{2\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij}(y_i - y_j)^2}{\sum_i (y_i - \bar{y})^2} \tag{4}$$

where n is the number of observations, w_{ij} is the spatial weight for a pair of observation i and j, y_i and y_j are raw values of observations i and j, respectively and $\bar{y}$ is the mean value of the whole dataset. Unlike Moran's I, which uses cross-products on the standartized values, Geary's C computes differences of raw values, which in case of very different neigbouring values, it spikes the resulting Geary C value up, making it more sensitive to appearance of negative SAC. Higher values of C indicate negative spatial correlation and values close to zero indicate positive spatial autocorrelation, so the values only have a lower bound (zero) and can reach any positive value. Values close to one indicate random configuartion of values.

Local Geary C, proposed by Anselin, can be written as

$$c_i = \sum_j w_{ij}(y_i - y_j)^2 \tag{5}$$

where n is the number of observations, w_{ij} is the spatial weight for a pair of observation i and j, y_i and y_j are raw values of observations i and j, respectively and $\bar{y}$ is the mean value. High values of local c_i highlight areas of abrupt changes in value, but as any LISA, its statistical significance that identifies wherther it actually matters.

2.1.2.3 Gettis Ord G

Proposed by Getis and Ord (1992), the global G is a measure of spatial autocorrelation based on distance, unlike Morans I or Geary C, which uses contiguity weights in vast majority of cases. It can be written as

$$G(d) = \frac{\sum_i \sum_j w_{ij}(d) y_i y_j}{\sum_i \sum_j y_i y_j} \tag{6}$$

where $w_{ij}(d)$ is the binary weight between observation i and j based on a distance band (1 if within the distance band, 0 if not). The weights don't have to be strictly binary, but usually they are. G measures the degree of positive spatial autocorrelation. The local version of G can be defined in two ways, G_i doenst include observation's own value, but G_i^* does, therefore

$$G_i = \frac{\sum_{j \neq i} w_{ij} x_j}{\sum_{j \neq i} x_j}, G_i^* = \frac{\sum_j w_{ij} x_j}{\sum_j x_j} \tag{7}$$

G_i and G_i^* help identify hot spots (areas with high values) and cold spots (

2.1.3 Regionalization

Regionalization is the process of creating regions (Duque et al., 2007). It is essentially grouping similar observations together, which we recognize as clustering, but with a spatial constraint that two observations (polygons) can be in the same cluster only if they share a border - contiguity. Regionalization is therefore known as spatially constrained clustering.

The objective of any clustering method is to minimize the dissimilarity within each cluster. The overall dissimilarity of a dataset is described with the sum of all pairwise distances T:

$$T = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} d_{ij} \tag{8}$$

where d_{ij} is a measure of dissimilarity. T consists of two complementary components: within-cluster dissimilarity W and between-cluster dissimilarity B, which can be evaluated during the clustering (regionalization) process:

$$T = W + B \tag{9}$$

where

$$W = \frac{1}{2}\sum_{h=1}^{k}\sum_{i\in h}\sum_{j\in h} d_{ij}, \qquad B = \frac{1}{2}\sum_{h=1}^{k}\sum_{i\in h}\sum_{j\notin h} d_{ij} \tag{10}$$

W serves as a loss function, the goal is to minimize it. Some methods explicitly use W as the clustering criterion (SCHC using Ward's linkage) whereas others use other metrics to approximate the goal.

This chapter will serve as an overview of widely used methods and in the experimental part, we will test the robustness to polygonal coverage violations of two methods - agglomerative clustering/SCHC and SKATER.

2.1.3.1 Agglomerative clustering/SCHC

Spatially constrained hierarchical clustering is a bottom-up approach, where each observation starts as its own cluster ($k = n$) and the clusters (or single observations) are being recursively merged based on their dissimilarity. Dissimilarity is computed based on definition of cluster distance in attribute space, referred to as linkage. By distance, we mean the euclidean one as it is used in vast majority of applications and also required for the use of Ward's linkage, which is preferred for regionalization and will be further described in this chapter. Other linkages are single linkage, using the minimum distance between clusters, complete linkage using the maximum distance and average linkage using the mean distance.

The dissimilarity is denoted in dissimilarity matrix D, which is initially nÖn as each observation starts as its own cluster, and each element d_{ij} represents the dissimilarity between each pair of clusters (or observations). The algorithm then merges the two clusters/observations, that have the smallest dissimilarity (smallest value from D) under the contiguity constraint. D is then updated using an updating formula, which computes the dissimilarities between the new cluster C and a cluster P, reducing both dimensions of D by 1 in each iteration.

Ward's method, developed by Ward (1963), explicitly tries to minimize the within-cluster dissimilarity (variance) defined trough the sum of squared errors (WSS). The input structure to use Ward's method is a nÖp matrix X, where n is the number of observations, p is the number of variable and each element represents the (standartized) value of each variable for each observation. WSS can be then formally defined as:

$$WSS = \sum_{i \in C} (x_i - \bar{x}_C)^2 \tag{11}$$

where x_i is a single row of X (the vector of attributes for observation i) and $\bar{x}_C$ is the centroid of all observations in cluster C. Intuitively, the overall WSS increases when we merge clusters, so the goal is to minimize this increase. This actually turns out to be merging clusters/observations which's centroids are closest to each other based on a dissimilarity matrix D. Ward's method expresses the dissimilarity trough the square of the Euclidean distance, therefore the dissimilarity between clusters A and B:

$$d_{AB}^2 = \frac{2n_A n_B}{n_A + n_B} \|\bar{x}_A - \bar{x}_B\|^2 \tag{12}$$

The updating formula to compute the dissimilarities between the new cluster C and a point (cluster) P after merging clusters A and B is defined as:

$$d_{PC}^2 = \frac{n_A + n_P}{n_C + n_P} d_{PA}^2 + \frac{n_B + n_P}{n_C + n_P} d_{PB}^2 - \frac{n_P}{n_C + n_P} d_{AB}^2 \tag{13}$$

As mentioned above, the clusters/observations can be only merged under the contiguity constraint, which is denoted in spatial weight matrix W, meaning that observations i and j can be merged only if $w_{ij} \neq 0$. AFter the merge, W needs to be updated. If i and j were merged into a new cluster C, i-th and j-th row and column will be replaced by C. The relationship between C and the rest of the clusters P will be denoted following $w_{PC} = 1$ if $w_{iP} \neq 0 \vee w_{jP} \neq 0$.

Common pratice is to define the number of clusters or number of regions in case of regionalization. This parameter stops the the hierarchical tree building called a dendrogram (see fig. x) but software tools like sklearn allow to built the whole tree even though it wont be necessary for the clustering.

-dendrogram

-definovany jen pro 1 componentu?

-nevratny

2.1.3.2 SKATER

Spatial 'K'luster Analysis by Tree Edge Removal (SKATER), developed by AssunÇão et al. (2006), is another hierarchical regionalization method but unlike SCHC, it is not agglomerative, but divisive. It is based on pruning a minimum spanning tree (MST). MST is a graph structure, defined as subset of edges that connect all nodes of the graph with minimum possible total edge weight.

MST for the purpose of SKATER algorithm is a subset of edges from contiguity spatial weights graph (see def in chapter x). The weights are assigned to each edge usually based on their Euclidean distance in attribute space, therefore:

$$d(i,j) = d(x_i, x_j) = \sum_{l=1}^{n} (\mathbf{x}_{il} - \mathbf{x}_{jl})^2 \tag{14}$$

where x_i, x_j are attribute vectors of observations i and j, which is understood to be the dissimilarity.

The initial MST has exactly $n-1$ edges. All observation are considered to be in one region. The MST is pruned by removing an edge which increases the between-cluster dissimilarity the most. This is accessed trough calculating the sum of squared devations (SSD) calculated as ... for each subtree and subtrating them from the SSD of the whole tree. If we are about to split a tree into subtree A and B, the increase of between cluster dissimilarity is calculated as

$$\Delta SSD = SSD_T - (SSD_A + SSD_B) \tag{15}$$

and needs to be the largest.

2.2 Topology in geography

Topology is a mathematical discipline that “studies properties of spaces that are invariant to continuous deformation” (University of Waterloo (2024)). In geographic context, it means focusing strictly on the spatial relationships between features, such as connectivity, adjacency, and containment (citace), rather than their absolute geometric shape, distance, or magnitude. The discipline was formally established in the early 20th century, however its roots trace back to Leonhard Euler's solution to the Seven Bridges of Königsberg problem (Euler (1741)). The objective of the problem was to cross every bridge in the city exactly once and return to the starting point. The physical distances and exact shapes of the landmasses were entirely irrelevant to the solution, the outcome depended solely on knowing which bridges connected which landmasses. By abstracting the city into a graph (see definition in chap. xx) while preserving the information about the connectivity of the landmasses, he was able to solve the problem.

-fig konigsberg bridges a graph

Probably one the most important connections between topology and geography is the Four Colour Theorem, proposed by Francis Guthrie 1852. It states that any planar map consisting of polygons can be coloured using only four colours without any pair of neigbouring polygons sharing the same colour. The theorem was proven more than a hundred years later by Appel and Haken (1977) and it relied entirely on topological principles, specifically, polygonal adjacency. Similarly to the Königsberg bridges, solving the problem required abstraction into a graph, a mathematical structure frequently used to describe spatial and topological relationships.

The formal study topological relationships is rooted in point-set topology, also known as general topology, as it is the foundation to many other related disciplines. It relies on the concepts set theory, introduced by Cantor (1874), like set operations (e. g. intersection, union, complement), as well as the fundamental notions of subsets, open sets, and neighborhoods. Hausdorff (1914) formalized topological space in his book on set theory, introducing the foundations of point-set topology.

A topological space is a pair (X, τ), where X is a set of points and τ is a collection of subsets of X (the topology). τ must satisfy the following axioms:

1. Any union of elements of τ belongs to τ.
2. Any finite intersection of elements of τ belongs to τ.
3. X and $\emptyset$ belong to τ.

Based on the definition of topological space, Hausdorff (1914) formally defined the interior, boundary and exterior of a set. These three topological properties are the core of topological models used in spatial analysis. Given a set Y in topological space (X, tau):

- interior (doplnim)
- boundary
- exterior

-Identifying these relationships is essential in numerous geographical disciplines, such as landscape ecology, hydrology, geography of transport, cartography and spatial statistics (Papadimitriou (2023)).

2.2.1 Topological models

During the 1980s, terms for spatial relations were mentioned in literature, however, they lacked formal definitions and were mostly treated as axiomatic (Egenhofer and Franzosa (1991)). The first rigorous framework for topological relationships was the *4-Intersection Model* (*4-IM*) developed by Egenhofer and Franzosa (1991). It applied interior and boundary definitions from point set topology to a geospatial context. The *4-IM* served as the predecessor to the *Dimensionally Extended 9-Intersection Model* (*DE9-IM*) (Clementini et al. (1993)), which is now the standard adopted by the *Open Geospatial Consorcium* (*OGC*). Expanding the *4-IM*, it is based on a 3×3 matrix of all possible intersections between interior, boundary and exterior of two geometries (point, line or polygon) and further describes the dimension of those intersections:

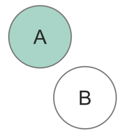
- False (-1) if there is no intersection,
- 0 if the intersection is a point,
- 1 if the intersection is a line,
- 2 if the intersection is a polygon.

-tab de9im matrix, nebo fig?

In most use cases, non-empty intersections (0, 1 or 2) can be universally treated as *True* when the result is invariant to the intersection dimension. However, for defining contiguities the dimension of the intersection is required (see chapt. xx). Specific value combinations in the matrix have standard names, which are referred to as spatial predicates. The *C++*

library *GEOS* - a core dependency of many libraries (e. g. *Shapely*, *Geopandas*), databases (e. g. *PostGIS*), and applications (e. g. *QGIS*) - recognizes 9 spatial predicates (see tab **?**) GEOS contributors (2025). For computational purposes, the matrix is often flattened into a 9-character DE-9IM string.

Table 1: Spatial predicates. - tohle taky dopnim

Spatial predicate	DE-9IM string	Description	Illustration?
Intersects	??		
Touches	FT*******		
	F**T*****		
	F***T****		
			
Disjoint	FF*FF****		
Crosses	T*T******		
	0********		
Within	T*F**F***		
Contains	??		
Overlaps	T*T***T**		
Equals	TFFFTFFFT		
Covers	??		

Software libraries implement these predicates alongside additional ones as boolean functions, such as modified formulations (e. g. `contains_properly` in *Shapely* library) or inverse relationships (e. g. `covered_by` in *Shapely*). Custom spatial queries can be formalized by defining the flattened DE9-IM string (e. g. `relate` in *Shapely*, `ST_relate` in *PostGIS*). Topological relationships derived from *DE-9IM* are the underlying framework for many spatial operations including overlay analysis, contiguity detection, and topology validation.

2.3 Topological validity and polygonal coverage

Spatial data often suffers from errors from a variety of sources. These inaccuracies are typically introduced during manual vectorization or data entry, data transfer, geometric editing operations (such as simplification or reprojection), or the integration of heterogeneous datasets of varying ages (Spatial Eye (2025), GIS Geography (2024), Servigne et al. (2000)).

Servigne et al. (2000) categorizes spatial consistency errors into three primary types:

- Structural errors: Arising when the chosen data structure cannot adequately accommodate the data model.
- Geometric errors: Conceptually invalid geometry (e. g. self-intersecting polygons).
- Topo-semantic errors: Occurring when the geometric representation violates real-world logical constraints (e.g., a single building footprint incorrectly intersecting two distinct parcels).

Geometric and topo-semantic errors are particularly problematic when detecting topological relationships, as they are not immediately noticable. The most commonly used data structures in modern GIS are "spaghetti" models (e.g., GeoJSON, Shapefile, Parquet, and GeoPackage). These models store each geometry as an isolated record with no information about their mutual relationships, therefore they need to be derived from the geometry (vertices). In presence of geometrical inaccuracies, accurate detection of topological relationships is impossible.

According to Ubeda and Egenhofer (1997), the topology error correction process consists of three steps: definition, checking, and correction. Definitions of the errors is esentially establishing the spatial constraints input data must satisfy. These user-defined constraints are derived from formal spatial predicates. Many of these rules are explicitely implemented as ready-to-use topology checkers in desktop GIS software like *QGIS* and *ArcGIS Pro* (e. g. `must not have dangles` for line features, `must not overlap` for line or polygonal features).

Polygonal coverage, or also planar enforcement, is a specific case of these rules. It is a strict requirement for contiguity detection Rey et al. (2022). Valid polygonal coverage requires specific set of topological constraints, which it must satisfy Jts (2022):

- polygon interiors must not overlap (no slivers),
- a shared boundary between polygons must have the same set of vertices (no gaps).

-fig topology errors slivers, gaps

-Geoplanar also mentions non-planar touches and edges, which by the general definition meet polygonal coverage (planar enforcement) requirements, but geometrically, they would violate it and contiguity detection would be compromised. Enforcing valid polygonal coverage ensures accurate spatial analysis and improves cartographic visualization.

The contiguity definitions seem intuitive, however, the definitions of sharing an edge or sharing a vertex are incomplete. In libpysal, sharing an edge is implemented as sharing two consecutive vertices and similarly, sharing a vertex means that both polygons have a vertex of the same coordinates. This can become problematic in case of non-planar edges and non-planar touches (see Fig.) (citace geoplanar). In situations like these, Queen contiguity wont be detected as the polygons are not planar enforced.

-Libpysal implements Queen and Rook contiguity, alongside fuzzy contiguity, which serves as more flexible

Various software tools can evaluate the datasets for topology violations. *Shapely*, python wrapper for the *GEOS* library, implements functions for checking individual geometric validity rather than global topology. *JTS Topology Suite* (*JTS*), an open source Java software library, can evaluate an entire polygonal coverage, returning error strings with the specific coordinates of violations. Desktop software like *QGIS* and *ArcGIS Pro* evaluate the selected topology rules (i. e. for polygonal coverage `must not overlap` and `must not have gaps`) and visually highlight every error on the map.

While *JTS*, *QGIS*, and *ArcGIS* Pro validate topology by checking individual geometries, Mapshaper temporarily converts the spaghetti model into an explicit topological data structure. Instead of storing independent geometries in rows, Mapshaper decomposes a polygonal dataset into an arc-node graph: creating edges (arcs) for boundaries and nodes where three or more polygons intersect. Polygons are then defined mathematically through sequences of these shared arcs, eliminating coordinate redundancy. In this explicit topological structure, the information detailing which polygons neighbor each other is inherently stored within the data model itself.

-Fixing errors can be either done (semi)automatically or by hand, involving a snapping treshold... a zapomnela jsem na geoplanar :(

-even though such tools exist, topological errors still occur, a presne proto pisu tuhle praci

3 Methodology

3.1 Simulated data

Three sizes of polygonal datasets were generated (n = 25, 100, and 400). To vary the average number of neighbours, three lattice types (square, pentagonal, and hexagonal) were applied to each size, yielding 9 distinct spatial configurations (fig x).

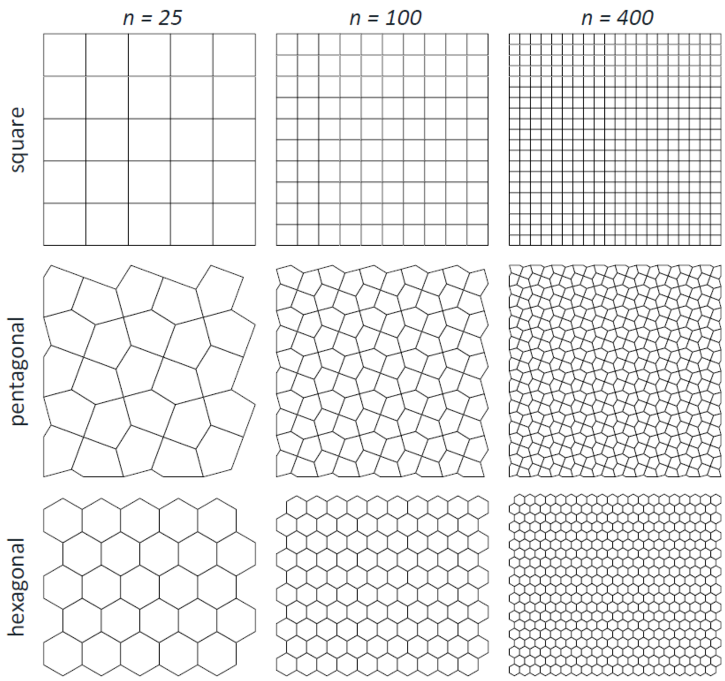


Figure 1: Simulated data spatial configurations.

3.1.1 Spatial autocorrelation

Attribute values were generated trough a pure spatial process using a spatial autoregressive (SAR) model, similarly to Florax and Rey (1995), calculated by the following equation for each of the nine spatial configurations:

$$y = (I - \rho W)^{-1}\epsilon$$

where y is the resulting vector of generated attribute values, I is a identity matrix, ρ is the spatial autoregressive parameter, W is Rook contiguity spatial weight matrix derived from each spatial configuration and ϵ is random normal error vector. ρ was varied from -0.9 to 0.9 in increments of 0.1, with 10 iterations per value, see examples in Fig. x. This produced 190 unique value datasets per spatial structure representing a wide spectrum differently spatially autocorrelated datasets.

-fig ruzny spatial autocorrelations

3.1.2 Regionalization

First, regions were predefined on all three lattice types at two dataset sizes, $n = 100$ and $n = 400$. For each of those six spatial configurations, 5 polygons were randomly selected as seeds, which grow into spatially connected regions (Guo et al., 2023), repeated 10 times, generating 10 zonations for each spatial configuration. For $n = 100$, the number of minimum polygons is set 12 and for $n = 400$ to 48.

Afterwards, the data was generated to exhibit strict stratified heterogeneity, meaning that true regression coefficients are constant within each predefined region but differ across regions. The underlying data-generating process follows a simple linear model used in Guo et al. (2023):

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

where the independent variables are drawn i.i.d. from a uniform distribution $x_1, x_2 \sim \mathcal{U}[0, 1)$ and the error term ϵ is from is from normal distribution with mean 0 and variance σ^2, therefore $\epsilon \sim \mathcal{N}(0, \sigma^2)$. $\sigma = 0.1$ for all simulations.

For each of the 5 regions, covariate coefficients β_1, β_2 take different values from $\{-2, -1, 0, 1, 2\}$ and $\beta_0 = 0$ in all regions. For each zonation, 5 attribute values were generated by this process, which were then fitted into regionalization models.

3.2 Real data

3.3 Contiguity corruption

To simulate erroneous topology, controlled adjacency corruptions are applied to a set of topologically correct polygons. These corruptions consist of removing edges from the W

reflecting Rook contiguity of polygons. Missing adjacencies are spatially distributed in three ways:

- randomly, where adjacency relations are removed uniformly at random across the entire graph,
- on region borders, simulating errors arising from the integration of multiple spatial datasets,
- locally, where adjacencies are removed within a spatially contiguous subregion, representing localised human digitisation or editing errors.

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